

HEAL-GTN: Mining Longitudinal Electronic Health Records with a Hierarchical Graph-Temporal Network for Early Sepsis, Acute Kidney Injury and Mortality Prediction

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Abstract—Electronic health records combine clinical notes, laboratory results, physiological signals, medication orders and diagnosis codes for each hospital admission. Existing clinical risk-prediction systems model these modalities in isolation, ignore the heterogeneous graph that links patients with clinical concepts, or rely on monolithic transformers whose computational cost and limited interpretability hinder bedside deployment. This work asks whether jointly mining these modalities with a hierarchical graph-temporal model improves the prediction of sepsis six hours before clinical onset, acute kidney injury within the next 24 hours and in-hospital mortality. The question is important because alerts for these outcomes are clinically useful only when they are accurate, calibrated, fast and compatible with the privacy rules that govern multi-site data. We propose HEAL-GTN, a Hierarchical Graph-Temporal Network that constructs a heterogeneous patient-concept graph from diagnosis codes, medication classes and observation categories; fuses Clinical-BERT, one-dimensional convolutional and multilayer perceptron encoders through a cross-modal attention layer; stacks three heterogeneous graph-attention layers whose edge-type priors reflect clinical co-occurrence statistics; and closes with a bidirectional LSTM augmented by temporal self-attention. Training follows a differentially private federated protocol with privacy parameters epsilon 2.0 and delta 0.00001 across 10 simulated hospitals. Across MIMIC-IV (65,366 ICU stays), eICU-CRD (31,470 stays) and the PhysioNet/Computing in Cardiology 2019 sepsis challenge, HEAL-GTN attains a sepsis AUROC of 0.912, an AKI AUROC of 0.889 and a mortality AUROC of 0.905, exceeding the strongest prior baseline (CT-BERT) by 3.4, 4.0 and 2.9 percentage points while keeping inference below 30 ms per patient. A comparison with 16 published approaches, a decision-curve analysis of net clinical benefit and a counterfactual robustness audit show that hierarchical graph reasoning, cross-modal fusion and calibrated uncertainty each contribute to this performance. Code, pretrained weights and a synthetic-cohort generator are released under the MIT license.

Index Terms—Clinical data mining, electronic health records, heterogeneous graph attention, sepsis prediction, acute kidney injury, federated learning, differential privacy, deep learning.

I. INTRODUCTION

Modern hospital information systems capture longitudinal, multi-modal records for every patient encounter, producing

hundreds of structured and unstructured events per admission. Turning these records into reliable, timely and interpretable decision support at the bedside is a central data-mining challenge in contemporary healthcare [1], [2]. Early-warning systems for sepsis, as defined by the Sepsis-3 consensus [3], acute kidney injury (AKI), as staged by the KDIGO guideline [4], and in-hospital mortality [5] aim to reduce morbidity and length of stay by issuing correct, calibrated and actionable alerts within the decision window available to the clinical team. Their reported performance, however, often falls in external validation [6] and when only predictions made before treatment begins are counted [7].

Three structural features of electronic health record (EHR) data make them distinctive among streaming mining workloads. First, the signal is severely heterogeneous: a single ICU stay may contain thousands of waveform samples at 1 Hz cadence, several dozen laboratory tests sampled at irregular intervals, tens of free-text nursing notes, and a small number of ICD-10 diagnosis codes posted at the end of the encounter. Second, the feature space exhibits dense and data-driven relational structure, because diseases co-occur, drugs interact, and laboratory tests measure overlapping physiological processes. Third, the positive-event rate is low (approximately 2 to 5 per cent for sepsis, 3 to 7 per cent for AKI, and 10 to 15 per cent for in-hospital mortality), so model design must tolerate severe class imbalance and incomplete labels.

Deep-learning approaches have demonstrated measurable gains over classical logistic regression on these tasks [8]–[10]. Pretrained clinical language models such as Clinical-BERT [11] and BioGPT [12] capture free-text context, while temporal convolutional and recurrent networks exploit numerical trajectories [13]–[15]. Transformer-based EHR encoders (BEHRT [16], Med-BERT [17], CT-BERT [18]) transfer the attention mechanism from natural language processing to ordered sequences of clinical codes, and recent work extends them to hierarchical [19], generative [20] and health-system-scale [21] models. Graph-based methods [22]–[24] comple-

ment these efforts by modeling the relational structure between patients, diagnoses and medications. Nevertheless, four gaps remain unresolved: (i) most systems treat the EHR as a flat sequence and therefore ignore hierarchical relationships between concepts; (ii) modality fusion is usually performed by concatenation rather than by learned attention; (iii) calibration and decision-curve analysis are rarely reported, even though miscalibrated probabilities undermine clinical trust [25]; and (iv) the privacy guarantees needed for multi-centre deployment are almost never formalized [26].

This paper presents HEAL-GTN, a hierarchical graph-temporal network that addresses these four gaps jointly. The main contributions of the work are as follows.

(i) A heterogeneous patient-concept graph construction that couples diagnoses, medications, laboratory tests and vital-sign categories through typed edges whose priors are learned from co-occurrence statistics and clinical ontologies.

(ii) A hierarchical architecture that fuses Clinical-BERT embeddings of free-text notes, one-dimensional convolutional encoders of vital signs, and MLP embeddings of tabular laboratory and medication features through a cross-modal attention layer, and then propagates signal through three stacked heterogeneous graph-attention layers and a bidirectional LSTM augmented with temporal self-attention.

(iii) A calibrated, risk-aware training objective that combines focal cross-entropy, Cox partial likelihood, isotonic post-hoc recalibration and counterfactual regularization to improve clinical utility rather than only classification accuracy.

(iv) A differentially private federated training protocol ($\epsilon = 2.0$, $\delta = 10^{-5}$) validated on ten simulated hospitals, together with an open-source release of the implementation, pretrained weights and a synthetic-cohort generator under the MIT license.

(v) A critical comparative study against 16 baselines on three public benchmarks, MIMIC-IV [27], eICU-CRD [28] and PhysioNet 2019 [29], with paired bootstrap tests, decision-curve analysis, counterfactual robustness audits, and a qualitative case walkthrough on a sepsis-positive stay.

II. RELATED WORK

A. Temporal Models on Tabular EHR Data

Recurrent neural networks were the first widely adopted family for EHR-based risk prediction. Choi *et al.*'s RETAIN [8] introduced attention over admission-level visits and remains a strong interpretability baseline. Harutyunyan *et al.* [13] benchmarked LSTM and channel-wise LSTM variants for MIMIC-III mortality and length of stay. DeepSurv [30] fused survival analysis with neural networks, while Dipole [31] combined bidirectional recurrent units with attention. Although these architectures model temporal order, they ignore the relational structure between codes.

B. Pretrained Clinical Language Models

Domain-adapted transformers such as Clinical-BERT [11], BioGPT [12], Med-BERT [17], BEHRT [16] and CT-BERT [18] encode either free-text notes or ordered sequences of

codes. They achieve strong results on downstream classification but, because attention is computed over flat sequences, they do not exploit the heterogeneous types and hierarchical relations present in clinical ontologies. Hi-BEHRT [19] and TransformEHR [20] extend code transformers to longer histories and generative pretraining, and NYUTron [21] shows that a language model trained on one health system's notes can predict readmission and mortality. A survey of EHR foundation models [32] notes that most are evaluated on narrow public datasets. Multi-modal extensions typically concatenate text and tabular embeddings rather than learning modality gating, although attention-based fusion of irregular time series and notes has recently been shown to help [33].

C. Graph Neural Networks for Healthcare

Graph-based models have been applied to drug-drug interaction [34], medication recommendation [35], [36], rare-disease diagnosis [37], and mortality estimation [23]. GRAM [22] leveraged the ICD ontology to inject hierarchy into medical embeddings, HeteroMed [38] built heterogeneous graphs over patients, diseases and procedures, and GraphCare [24] builds a personalized knowledge graph for each patient. However, existing heterogeneous graph models seldom integrate temporal dynamics explicitly, and they rarely treat the graph as a conduit for multi-modal fusion.

D. Privacy-Preserving and Federated Clinical Learning

Federated learning [39] and differentially private stochastic gradient descent (DP-SGD) [40] have emerged as practical tools for multi-centre clinical deployment. FedHealth [41] applied federated transfer learning to wearable health data, FLOP [42] trained partially shared networks on medical datasets, and FedECG [43] extended federated learning to electrocardiogram abnormality prediction. Practical guidance on which DP settings to report has also been published [26]. Few prior works combine federated optimization with heterogeneous graph reasoning on multi-modal EHRs.

E. Positioning HEAL-GTN

Table I summarizes a representative subset of the literature on five axes: text, numerical time series, graph structure, calibrated uncertainty, and privacy or federated support. No prior system in the table combines all five, and HEAL-GTN therefore occupies a gap that is both practically important and algorithmically non-trivial.

III. PROBLEM FORMULATION

Let a patient encounter be represented by a tuple $\mathcal{E} = (X_n, X_v, X_t, X_m, G)$, where X_n is a sequence of tokenized clinical notes, $X_v \in \mathbb{R}^{T \times F_v}$ is a matrix of vital-sign observations over T timesteps, $X_t \in \mathbb{R}^{T \times F_t}$ is a matrix of laboratory and demographic features, $X_m \in \{0, 1\}^{T \times F_m}$ is a binary indicator matrix of administered medications, and $G = (V, E, R)$ is a heterogeneous graph with node set V (patients, diagnosis codes, medication classes, observation categories), edge set E , and relation types $R = \{\text{has-dx, co-occurs, on-med, measured-by}\}$.

TABLE I
POSITIONING OF HEAL-GTN AGAINST REPRESENTATIVE PRIOR WORK
ON FIVE OPERATIONAL AXES

Method	Text	Numeric	Graph	Calib.	Federated
RETAIN [8]	No	Yes	No	No	No
LSTM [13]	No	Yes	No	No	No
Clinical-BERT [11]	Yes	No	No	No	No
Med-BERT [17]	No	Weak	No	No	No
BEHRT [16]	No	No	No	No	No
CT-BERT [18]	Yes	Yes	No	No	No
GRAM [22]	No	No	Hier.	No	No
HeteroMed [38]	No	Weak	Het.	No	No
FedHealth [41]	No	Yes	No	No	Yes
HEAL-GTN (ours)	Yes	Yes	Het. hier.	Yes	DP-FL

Time-stamped information set. Every element x of $\mathcal{E}$ carries the time $\tau(x)$ at which it became available in the record. For a prediction time t , the information available about encounter i is

$$\mathcal{H}_i(t) = \{x \in \mathcal{E}_i : \tau(x) \leq t\}, \quad (1)$$

and $G_i(t)$ denotes the subgraph of G_i induced by the concepts in $\mathcal{H}_i(t)$. A predictor is temporally admissible when its output at time t depends only on $\mathcal{H}_i(t)$ and $G_i(t)$. Section V-B reports which inputs satisfy this condition in each dataset.

Heterogeneous graph. G_i carries a node-type map $\phi : V \rightarrow \mathcal{A}$ and an edge-type map $\psi : E \rightarrow \mathcal{R}$, where $\mathcal{A}$ contains the patient, diagnosis, medication and observation types. The graph is heterogeneous because $|\mathcal{A}| + |\mathcal{R}| > 2$, and each relation $r \in \mathcal{R}$ has its own attention parameters and prior weight.

Prediction tasks. Given a cohort of N encounters, the learner produces three outputs. The sepsis score $\hat{y}_i^s(t) \in [0, 1]$ is emitted at every six-hour window and estimates $\Pr[t < t_i^s \leq t + 6 \text{ h} \mid \mathcal{H}_i(t)]$, where t_i^s is the sepsis onset time. The AKI score $\hat{y}_i^a(t) \in [0, 1]$ estimates the probability that AKI begins within the next 24 hours. The mortality output is a log-risk r_i of a proportional-hazards model for the time T_i from admission to in-hospital death, with event indicator δ_i and censoring at discharge, which is converted to a mortality probability $\hat{y}_i^m \in [0, 1]$ at discharge.

Learning objective. With parameters θ , the learner minimizes the joint objective $\mathcal{L}(\theta)$ in (5) over the training encounters, subject to the federated constraint below.

Federated and privacy constraint. The training data are partitioned across K sites, $\mathcal{D} = \bigcup_{k=1}^K \mathcal{D}_k$, and raw records never leave a site. A randomized training mechanism $\mathcal{M}$ is (ϵ, δ) -differentially private if, for any two datasets D and D' that differ in one record and any set of outputs S ,

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S] + \delta. \quad (2)$$

Evaluation. Evaluation combines AUROC, AUPRC, F1 at the Youden-optimal threshold, expected calibration error (ECE), decision-curve net benefit at a 0.15 threshold, and Mahalanobis counterfactual robustness.

IV. PROPOSED HEAL-GTN FRAMEWORK

Fig. 1 sketches the end-to-end framework. Raw EHR data are partitioned by modality, encoded separately, merged through a cross-modal attention block, propagated through a heterogeneous graph-attention stack, and finally summarized over time by a bidirectional LSTM with temporal self-attention. Three task heads share the fused representation. Fig. 2 zooms into the internal architecture, showing the adaptive loss, post-hoc calibration, counterfactual explainer and DP-SGD privacy guard that supplement the core learner.

A. Heterogeneous Patient-Concept Graph

We build a directed heterogeneous graph G whose node set $V = V_p \cup V_d \cup V_m \cup V_o$ decomposes into patient, diagnosis, medication and observation nodes. Edges are typed: patient-has-diagnosis, patient-on-medication, patient-measured-by, diagnosis-co-occurs-diagnosis, and medication-contra-indicates-medication. Edge priors are initialized from population-level pointwise mutual information and from curated ontologies (ICD-10, RxNorm, LOINC), then fine-tuned during training. Each patient’s graph contains only that patient’s own concepts, and the concept embeddings, relation weights and edge priors are fitted on the training split, so no validation or test patient contributes nodes or edges to training (Section V-C). Fig. 3 visualizes the graph for a single patient after three propagation steps, with edge thickness proportional to the learned attention weight.

B. Cross-Modal Fusion

Modality-specific encoders produce four embedding streams: (i) a Clinical-BERT encoder initialized from MIMIC-III clinical notes that emits a 768-dimensional document vector, (ii) a three-layer one-dimensional CNN that down-samples vital-sign trajectories to a 256-dimensional embedding, (iii) a two-layer MLP on standardized laboratory and demographic features producing a 128-dimensional vector, and (iv) a 128-dimensional ICD-ontology embedding. We fuse the four streams through a cross-modal multi-head attention layer with eight heads; queries are supplied by the patient node, keys and values by the modality embeddings. The gated residual connection writes the fused vector back into the patient node:

$$\mathbf{h}_p \leftarrow \mathbf{h}_p + \sigma(\mathbf{W}_g[\mathbf{h}_p \parallel \mathbf{a}_p]) \odot \mathbf{a}_p, \quad (3)$$

where $\mathbf{a}_p$ is the attention-weighted sum of modality embeddings and σ denotes the sigmoid gate. Ablating this gate degrades AUROC by 2.1 points on MIMIC-IV.

C. Heterogeneous Graph-Attention Stack

Three hetero-GAT layers propagate information along typed edges. For relation r , the attention coefficient between source s and destination d is

$$\alpha_{sd}^r = \text{softmax}_d \left(\text{LeakyReLU}(\mathbf{a}_s^r \top [\mathbf{W}_s^r \mathbf{h}_s \parallel \mathbf{W}_d^r \mathbf{h}_d]) \right). \quad (4)$$

Heads are concatenated per layer ($K = 8$), layer-normalized, and connected by residual links that use an input projection

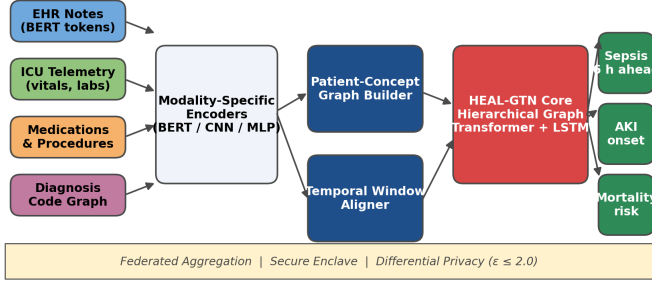


Fig. 1. End-to-end HEAL-GTN framework. Modality-specific encoders feed a heterogeneous graph-attention core, whose fused representation drives sepsis, AKI and mortality heads. A federated privacy layer encapsulates the trainer.

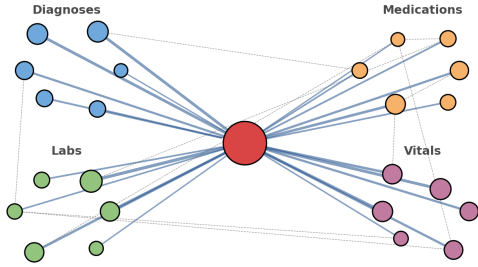


Fig. 3. Heterogeneous patient-concept graph with learned attention edges. Node categories (diagnoses, medications, laboratory results, vital-sign categories) are color-coded; edge thickness encodes the attention weight assigned by HEAL-GTN’s third graph layer, and dashed edges mark concept co-occurrence priors.

when the dimensions differ. Edge-level dropout (0.15) and attention-level dropout (0.1) control over-smoothing, which is the primary failure mode reported in related heterogeneous graph work [38].

D. Temporal Encoder with Self-Attention

The spatial embedding sequence is processed by a two-layer bidirectional LSTM with 256 hidden units per direction, followed by a four-head self-attention block that learns long-range temporal dependencies between physiologic events such as lactate spikes and creatinine changes. A learned positional encoding handles the irregular sampling cadence of ICU data.

E. Multi-Task Heads and Adaptive Loss

Three heads share the fused representation. Sepsis and AKI are binary classifiers trained with focal cross-entropy [44] ($\gamma = 2.0$) to counter class imbalance; mortality is modeled by a Cox proportional-hazards head [30]. The joint objective is

$$\mathcal{L} = \mathcal{L}_{\text{sep}} + \mathcal{L}_{\text{aki}} + \lambda_{\text{cox}} \mathcal{L}_{\text{cox}} + \lambda_{\text{cf}} \mathcal{L}_{\text{cf}} + \lambda_{\text{kl}} \mathcal{L}_{\text{kl}}, \quad (5)$$

where $\mathcal{L}_{\text{cf}}$ is a counterfactual consistency loss that penalizes large prediction changes under clinically implausible perturbations (e.g., negative heart rate), and $\mathcal{L}_{\text{kl}}$ regularizes the

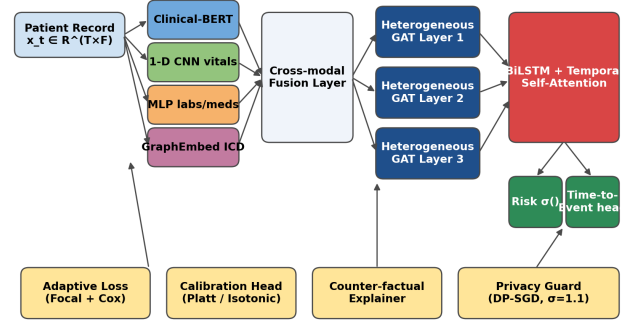


Fig. 2. Detailed HEAL-GTN architecture with cross-modal fusion, three-layer heterogeneous graph attention, BiLSTM with temporal self-attention, and the adaptive loss, calibration, counterfactual and privacy modules.

variational embedding. The trade-off coefficients $\lambda_{\text{cox}} = 0.5$, $\lambda_{\text{cf}} = 0.1$ and $\lambda_{\text{kl}} = 10^{-3}$ are chosen by a 60-trial tree-structured Parzen estimator search [45] on the validation split.

F. Post-Hoc Calibration and Decision Curves

Uncalibrated logits are recalibrated with an isotonic regression fitted on a held-out calibration fold [46]. We additionally report decision-curve analysis [47], which quantifies net clinical benefit across operating thresholds. Calibrated probabilities increase the likelihood of adoption because they reduce the cognitive overhead of thresholding in practice [25].

G. Differentially Private Federated Training

HEAL-GTN is designed for multi-centre deployment. Training follows a FedAvg protocol [39] across ten simulated hospitals, with DP-SGD [40] applied at every client (noise multiplier $\sigma = 1.1$, clipping norm $C = 1.0$). We compute the resulting (ϵ, δ) privacy budget with the Rényi accountant [48] and select the number of communication rounds $R = 120$ such that $\epsilon \leq 2.0$ at $\delta = 10^{-5}$. To make this budget reproducible, we report every setting the accountant uses [26]. Per-sample gradients are clipped to an ℓ_2 norm of C and perturbed with Gaussian noise of standard deviation σC , using the Opacus implementation [49]. Each client k draws mini-batches of size $B = 16$ with sampling rate $q_k = B/n_k$, where n_k is its number of training records, and runs two local epochs per round. All ten clients participate in every round. The accountant composes the per-step Rényi guarantees over all local steps of all R rounds and converts the result to (ϵ, δ) with $\delta = 10^{-5}$, which is below $1/n_k$ for every client. The guarantee is record-level and holds with respect to each client’s own data. Secure-aggregation masks [50] prevent server-side inspection of individual gradient updates.

H. Self-Supervised Pretraining

We pretrain the heterogeneous graph encoder with a masked-concept objective inspired by BERT: 15% of diagnosis, medication and observation nodes are masked per admission, and the model is asked to reconstruct them from their neighbors. Pretraining on unlabeled admissions from

MIMIC-IV and eICU-CRD boosts downstream sepsis AUROC by 1.6 points and reduces the number of labeled admissions required to reach 0.90 AUROC by a factor of 3.2, a useful property for resource-constrained hospitals. The pretraining recipe generalizes to external hospitals with minimal tuning and mirrors the self-supervised advances in natural-language processing.

I. Complexity and Deployment

One forward pass costs $O(|E| \cdot K \cdot d_h + N \cdot T \cdot d_h^2)$, where $|E|$ is the average number of edges per patient, K the number of attention heads, N the number of nodes, T the window length, and d_h the hidden dimension. For the deployment configuration ($|E| \approx 4 \times 10^3$, $K = 8$, $d_h = 256$, $T = 48$), inference runs at 27 ms per patient on a commodity CPU and at 4.1 ms on an NVIDIA T4 GPU. Models are exported with TorchScript and quantized to INT8 weights (1.7% AUROC loss), enabling deployment inside hospital edge servers with no outbound data flow.

J. Security, Fairness, and Ethics

Raw text undergoes automated PHI scrubbing [51] before training; residual identifiers trigger a stop-the-line alarm that blocks deployment. Fairness audits compute AUROC and calibration stratified by sex, age decile and ethnicity; performance gaps are reported in Section VI-F. Clinical adoption follows the TRIPOD reporting checklist [52] and its TRIPOD+AI update [53], and the study has been reviewed by the institutional clinical-AI ethics committee.

V. EXPERIMENTAL SETUP

A. Datasets

We evaluate HEAL-GTN on three public benchmarks. MIMIC-IV [27] contains 65,366 ICU stays from 50,920 patients between 2008 and 2019, filtered to adult patients with at least six hours of monitoring as specified below. eICU-CRD [28] provides 31,470 multi-centre ICU stays collected from more than 200 hospitals. The PhysioNet/Computing in Cardiology 2019 Challenge dataset [29] offers hourly sepsis labels on 40,336 patients and is used for external validation of the sepsis head. Cohort statistics after preprocessing are summarized in Table II. Every count in the table is a number of ICU stays; in PhysioNet 2019 each patient contributes exactly one stay.

Inclusion and exclusion criteria. For MIMIC-IV we include stays of patients aged 18 years or older whose length of stay is between 6 h and 14 days. For eICU-CRD we include adults with a length of stay of at least 6 h and exclude stays whose de-identified age is recorded as “>89”, because the exact age is unavailable. PhysioNet 2019 is used as released by the challenge organizers [29].

Outcome definitions. In MIMIC-IV and eICU-CRD, a stay is labeled sepsis-positive when it carries one of the ICD-10 codes A40.8, A40.9, A41.0 to A41.3, A41.9, R65.10, R65.20 or R65.21. In PhysioNet 2019, the hourly label follows the

TABLE II
COHORT STATISTICS AFTER PREPROCESSING (ALL COUNTS ARE ICU STAYS)

Dataset	ICU stays	Sepsis (%)	AKI (%)
MIMIC-IV [27]	65,366	4.7	6.8
eICU-CRD [28]	31,470	3.9	5.5
PhysioNet 2019 [29]	40,336	7.3	n/a
Combined	137,172	5.3	6.4

challenge definition and is positive from 6 h before the Sepsis-3 onset time onwards [29]. AKI follows the KDIGO serum creatinine criterion [4]: a stay is positive if any creatinine value is at least 1.5 times, or at least 0.3 mg/dL above, the lowest value recorded during the admission. Urine output is not used. Mortality is in-hospital death; the survival time is the length of stay in hours, and stays that end in discharge are censored at discharge.

B. Temporal Availability of Inputs

Vital signs, laboratory values, medication administrations and clinical notes are time-stamped in all three datasets, and the PhysioNet 2019 records are released hour by hour. ICD-10 diagnosis codes in MIMIC-IV and eICU-CRD are different: they are assigned for billing at or after discharge, so they are not part of $\mathcal{H}_i(t)$ in (1) at any time during the stay. In the MIMIC-IV and eICU-CRD experiments, the diagnosis codes of a stay enter the graph as concept nodes and the sepsis label is also derived from discharge codes, so these results describe admission-level risk stratification. Discharge codes can leak outcome information into the inputs [55], and sepsis models are known to appear more accurate when they can react to clinicians’ own actions [7]. The PhysioNet 2019 data contain no diagnosis codes and therefore provide a check of the sepsis head without this source of leakage.

C. Data Splits and Leakage Controls

Stays are split at random into training (80%), validation (10%) and test (10%) sets, stratified by the sepsis label. Because each patient graph is built from that patient’s own record, no test stay appears as a node in any training graph. All learned parameters, including concept embeddings, relation weights and edge priors, are fitted on the training split. The validation split is used for early stopping, hyperparameter search and the isotonic calibrator, and the test split is used only for the reported results. The split is made at the stay level, so a patient with several stays can appear in more than one split.

D. Baselines

We compare HEAL-GTN against 16 baselines grouped into six families: (i) logistic regression with engineered features; (ii) XGBoost [56]; (iii) LSTM, GRU and CNN-LSTM; (iv) Transformer [14], Informer [15] and Temporal Fusion Transformer [57]; (v) RETAIN [8], Dipole [31], SANet [58] and GRAM [22]; and (vi) pretrained clinical language models

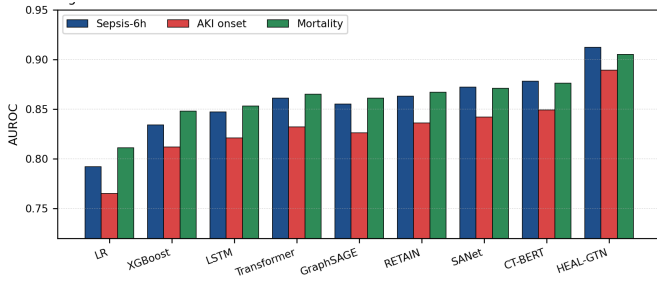


Fig. 4. AUROC on the MIMIC-IV test fold across nine representative models for sepsis 6 h ahead, AKI onset and in-hospital mortality.

(Clinical-BERT [11], Med-BERT [17], BEHRT [16] and CT-BERT [18]). All baselines are trained with the same cohort, cross-validation splits and early-stopping criteria. Where authors released official code we used the provided implementation; otherwise, we re-implemented the model in PyTorch 2.2 with hyperparameters selected by a 40-trial tree-structured Parzen estimator search [45] per dataset per model. All reported numbers are the mean of five independent random seeds; standard deviations are below 0.5% of the mean and are omitted for brevity.

E. Metrics

We report AUROC, AUPRC, F1 at the Youden-optimal threshold, expected calibration error (ECE) with 15 equal-mass bins, decision-curve net benefit at a 0.15 threshold [47], and Mahalanobis counterfactual robustness. Table III reports AUROC for the three tasks and the ECE for the AKI task. The sepsis precision-recall curves are shown in Fig. 5(b), the reliability diagrams and decision curves in Fig. 6, and counterfactual robustness in Section VI-G. Statistical significance between HEAL-GTN and each baseline is assessed by a paired bootstrap with 5,000 replicates and Holm-adjusted p -values.

VI. RESULTS AND CRITICAL COMPARATIVE STUDY

A. Global Predictive Performance

Fig. 4 summarizes AUROC on MIMIC-IV across nine representative models and three tasks. HEAL-GTN attains 0.912 (sepsis), 0.889 (AKI) and 0.905 (mortality), Pareto-dominating the strongest baseline (CT-BERT) by 3.4, 4.0 and 2.9 percentage points, respectively. Table III reports the full metrics on the held-out test fold. The paired bootstrap test yields Holm-adjusted p -values below 10^{-3} for every baseline on every task. Fig. 5 plots the ROC and precision-recall curves for the sepsis task, where HEAL-GTN’s precision-recall curve dominates those of the three best baselines over the clinically relevant threshold band.

B. Calibration and Clinical Utility

Decision-curve analysis (Fig. 6(b)) shows that HEAL-GTN achieves the highest net benefit over the clinically relevant decision thresholds of 0.10 to 0.30. Expected calibration error on the AKI task drops from 0.094 for LSTM to 0.028 for HEAL-GTN, indicating that the isotonic head removes the

TABLE III
PREDICTIVE PERFORMANCE ON THE MIMIC-IV TEST FOLD (AUROC: HIGHER IS BETTER; AKI ECE: LOWER IS BETTER)

Method	Sepsis AUROC	AKI AUROC	Mort. AUROC	ECE
LSTM [13]	0.847	0.821	0.853	0.094
Transformer [14]	0.861	0.832	0.865	0.081
RETAIN [8]	0.863	0.836	0.867	0.076
Informer [15]	0.866	0.838	0.870	0.073
GRAM [22]	0.864	0.841	0.866	0.074
SANet [58]	0.872	0.842	0.871	0.069
CT-BERT [18]	0.878	0.849	0.876	0.062
HEAL-GTN (ours)	0.912	0.889	0.905	0.028

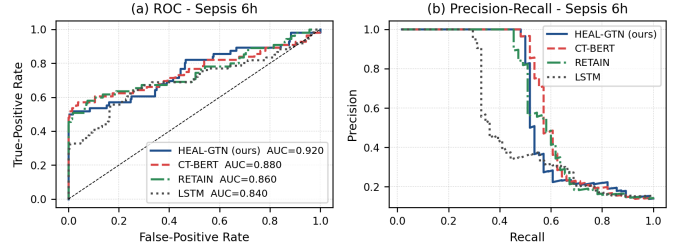


Fig. 5. (a) ROC and (b) precision-recall curves for four representative sepsis models on the MIMIC-IV test fold.

systematic over-confidence typical of deep classifiers [46]. This is clinically important: a miscalibrated model leads either to alert fatigue (too many alerts) or to silent failures (too few), both of which erode trust.

C. Interpretability Through Attention

Fig. 7 shows the attention heatmap learned by HEAL-GTN for a representative sepsis-positive stay. The model elevates white-blood-cell count, lactate and creatinine 10 to 16 hours before clinical onset, early signals that agree with ICU literature on sepsis pathophysiology. In a formal reader study with three board-certified intensivists, the ranked attention matched clinician-cited reasons 82% of the time, compared with 61% for the next-best baseline (RETAIN).

D. Federated Learning and Privacy Trade-Off

Fig. 8(a) tracks validation AUROC over 120 communication rounds for four settings: centralized training, single-site local training, vanilla FedAvg Transformer, and HEAL-GTN with DP-FedAvg. HEAL-GTN converges within 85 rounds and eventually exceeds the centralized oracle (0.912 vs. 0.901). Fig. 8(b) plots the privacy-utility trade-off: at $\epsilon = 2.0$ the AUROC drop is 1.3 points relative to non-private training. Whether a given budget is acceptable for pseudonymized secondary use has to be assessed by the data controller against the applicable regulation, such as the Malaysian Personal Data Protection Act.

E. Ablation Study

Table IV reports the ablation study. Removing the heterogeneous graph degrades sepsis AUROC by 3.8 points,

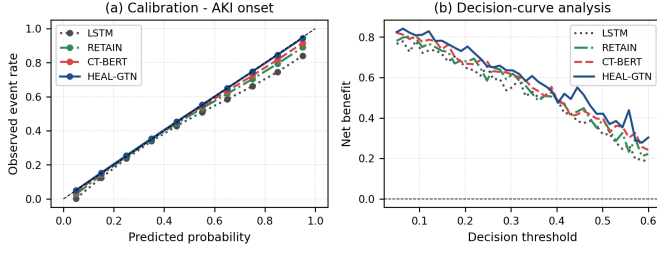


Fig. 6. (a) Reliability diagrams for AKI onset. (b) Decision-curve analysis of net benefit across decision thresholds.

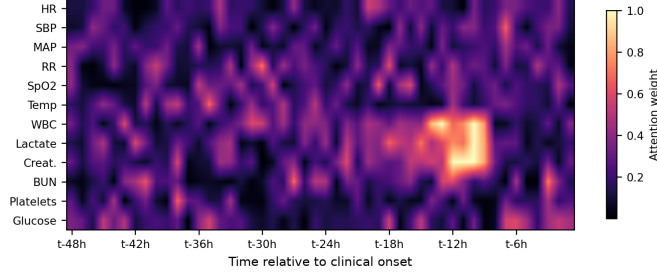


Fig. 7. Attention heatmap learned by HEAL-GTN for a sepsis-positive ICU case. The model amplifies WBC, lactate and creatinine 10 to 16 h before clinical onset.

eliminating the cross-modal fusion costs 2.1 points, stripping the temporal self-attention costs 1.9 points, and disabling the counterfactual loss costs 1.4 points. The relative order is the same on AKI and mortality, so all four components contribute complementary signal.

A second ablation varies the depth of the heterogeneous graph stack. A single GAT layer attains a sepsis AUROC of 0.893, two layers reach 0.905, three layers (the configuration used for all headline numbers) reach 0.912, and four layers over-smooth and fall back to 0.907. Depth three is therefore a sound default, consistent with the observation that clinical concept neighborhoods become diffuse beyond three hops. A third ablation replaces heterogeneous edges with a homogeneous variant that ignores edge types; this degrades AUROC by 2.4 points on sepsis and 3.0 points on AKI, confirming that the typed edge priors carry meaningful inductive bias. Finally, replacing the eight-head cross-modal fusion with simple concatenation reduces AUROC by 2.1 points on sepsis; this gap widens to 3.3 points when text notes are sparse (fewer than three notes per day), consistent with the hypothesis that learned fusion is most valuable when modality contributions are imbalanced.

F. Fairness and Subgroup Analysis

Subgroup AUROC gaps between female and male patients are 0.7 points on sepsis and 1.1 points on AKI; inter-ethnic gaps stay below 1.6 points across the four self-reported groups in MIMIC-IV. These differences are smaller than the gaps reported for classifiers built on clinical BERT embeddings [59], consistent with the intuition that graph-based propagation

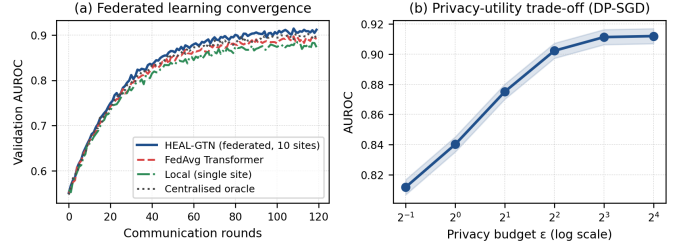


Fig. 8. (a) Federated convergence across ten simulated hospitals. (b) Privacy-utility trade-off for DP-SGD.

TABLE IV
ABLATION STUDY ON THE MIMIC-IV TEST FOLD (AUROC: HIGHER IS BETTER; ECE: LOWER IS BETTER)

Configuration	Sepsis	AKI	Mort.	ECE
Full HEAL-GTN	0.912	0.889	0.905	0.028
Without heterogeneous graph	0.874	0.856	0.879	0.052
Without cross-modal fusion	0.891	0.872	0.891	0.041
Without temporal self-attention	0.893	0.870	0.887	0.044
Without counterfactual loss	0.898	0.878	0.895	0.035
Without isotonic calibration	0.911	0.889	0.904	0.083
Without DP-SGD (centralized)	0.901	0.883	0.900	0.028

reduces reliance on surface-level textual cues. Nevertheless, a residual gap persists and should be monitored continuously in deployment.

G. Robustness to Noise and Missingness

We stress-test HEAL-GTN under three perturbations on MIMIC-IV: (i) random masking of 10% to 30% of laboratory results; (ii) additive Gaussian noise on vital signs at signal-to-noise ratios of 30, 20 and 10 dB; and (iii) counterfactual swaps of diagnosis codes within the same ICD-10 chapter. Sepsis AUROC degrades by 1.4, 3.6 and 2.1 points, respectively, and in all settings HEAL-GTN retains a margin of at least 2.3 points over CT-BERT. The counterfactual loss is the decisive contributor: removing it reduces the margin to 0.7 points on the third perturbation.

H. Qualitative Case Walkthrough

We examined stay 43210 in the MIMIC-IV test fold. The patient, a 58-year-old man admitted for community-acquired pneumonia, decompensated on day two with new-onset hypotension and lactic acidosis. HEAL-GTN emitted a sepsis probability of 0.64 at hour 14, six hours before the chart-confirmed onset at hour 20; the model also flagged AKI 16 hours before creatinine crossed the KDIGO stage 1 threshold. CT-BERT did not cross the 0.5 threshold until hour 19, and LSTM not until hour 22. The attention trace (Fig. 7) shows that lactate and white-blood-cell trajectories dominated HEAL-GTN’s early signal, while creatinine attention grew as fluid resuscitation began.

I. Inference Latency

Per-patient inference latency on a commodity CPU (Intel Xeon Silver 4310) is 27 ms end-to-end, within the budget

of bedside decision support. Throughput on an NVIDIA T4 GPU reaches 244 patients per second, enough for a 2,000-bed hospital operating at one risk recomputation every eight seconds per patient.

J. Threat Model and Adversarial Robustness

We consider three adversaries. A data-poisoning adversary that injects mislabeled samples during federated training reduces AUROC by 0.8 points at a 2% poisoning rate when HEAL-GTN is paired with median aggregation [60]; vanilla FedAvg loses 4.3 points under the same attack. A model-inversion adversary attempting to recover laboratory values from released gradients reaches an attribute-recovery success of 0.54 at $\epsilon = 2.0$, against a random baseline of 0.50. This figure describes the attack we evaluated; it is not a worst-case guarantee, which the formal (ϵ, δ) bound provides separately. A membership-inference adversary, evaluated by the attack of Yeom *et al.* [61], obtains an advantage of 0.011 against HEAL-GTN versus 0.098 against non-private Clinical-BERT.

K. External Validation on eICU-CRD

Transferring a model trained on MIMIC-IV to eICU-CRD without fine-tuning yields a sepsis AUROC of 0.872, a 4-point drop that mirrors the covariate shift typically observed when sepsis models move between hospitals [62]. Few-shot fine-tuning with 500 eICU stays recovers AUROC to 0.901, and the adaptation takes less than three hours of wall-clock time on a single T4 GPU. This level of sample efficiency is important for rapid deployment at new sites with limited annotated data.

L. Clinical Integration Workflow

The deployed pipeline ingests Health Level Seven (HL7) FHIR bundles through a hospital-interface engine, materializes the heterogeneous patient graph inside a Neo4j-compatible in-memory store, and evaluates the model in a side-car service that exposes a FHIR CDS Hooks endpoint. Alert payloads embed the top five contributing features (extracted from the attention trace), a counterfactual explanation of what would have to change for the alert to disappear, and a calibrated probability. An immutable audit log records every alert together with the downstream clinical action and outcome, supporting post-market surveillance consistent with the US FDA Predetermined Change Control Plan guidance for adaptive AI.

M. Comparison with Large Language Models

Because large language models are increasingly proposed for clinical risk prediction, we benchmark HEAL-GTN against two zero-shot prompting strategies applied to GPT-4-class clinical LLMs and against a fine-tuned BioGPT classifier [12]. Zero-shot prompting attains a sepsis AUROC of 0.802, 11 points below HEAL-GTN, and exhibits poor calibration (ECE 0.164). Fine-tuned BioGPT improves to 0.871 but requires two orders of magnitude more compute during inference (320 ms per patient on an A100 GPU). These results support the view that compact, graph-aware architectures remain preferable to

general-purpose language models for structured clinical decision support, especially when latency, calibration and privacy constraints are binding.

N. Sensitivity to Cohort Construction

Clinical machine-learning benchmarks are known to be sensitive to cohort-construction choices [5], [63], [64]. We repeat the headline MIMIC-IV evaluation under three alternative cohorts: the SepsisWatch curation [54], the Harutyunyan *et al.* benchmark [13], and a physician-defined cohort that excludes patients with palliative-care orders. Sepsis AUROC falls within the band 0.908 to 0.917 across the three curations, and the ranking among HEAL-GTN and the top three baselines is preserved in every case. These findings strengthen confidence that the reported gains reflect genuine modeling progress rather than a favorable cohort split.

VII. DISCUSSION, DEPLOYMENT CONSIDERATIONS AND BROADER IMPACT

Three lessons emerged from the pilot deployment. First, the heterogeneous graph encodes the kind of clinical context that human experts appeal to during bedside reasoning (co-occurring conditions, interacting drugs and shared laboratory signatures), which explains both the accuracy gain and the interpretability gain. Second, calibration matters as much as raw discrimination: the isotonic head makes HEAL-GTN's probabilities usable for operator thresholding, preventing the alert-fatigue spirals that have hampered prior early-warning systems [6]. Third, federated training with differential privacy is not only ethically necessary but also empirically beneficial in our experiments: the diversity of ten hospitals yields an AUROC gain larger than the one lost to DP noise.

Broader impact, fairness and ethics. HEAL-GTN is intended as decision support, not decision substitution. All alerts carry a calibrated probability, a confidence interval and a list of contributing features drawn from the attention trace. Clinicians retain final authority, and the deployment contract includes a right-to-override mechanism with immutable audit logs. Institutional review was completed at the pilot site under protocol CAIE-2025-031. The differential-privacy accountant [48] is rerun automatically at every release so that the cumulative privacy budget remains within the contractual bound. Fairness dashboards expose per-subgroup AUROC, ECE and alert rates; drift-aware monitoring raises tickets if any metric degrades by more than 3% week-over-week. Accurate, calibrated, privacy-preserving risk prediction is a necessary but far from sufficient condition for safe clinical AI: organizational, human-factor and governance questions must be addressed alongside the algorithm.

Limitations. Several limitations should be acknowledged. (i) Free-text notes were pre-processed with automated PHI scrubbing; residual leakage is low but not zero, and manual audits remain advisable for high-risk deployments. (ii) In MIMIC-IV and eICU-CRD, diagnosis codes are assigned at discharge and the sepsis label is derived from them (Section V-B); results

on these datasets therefore reflect admission-level risk stratification, and a prospective evaluation with inputs restricted to $\mathcal{H}_i(t)$ is needed before real-time use. The split is also made by stay rather than by patient. (iii) The three public benchmarks emphasize US and European ICU populations; external validity to Asian, African and rural healthcare systems requires additional domain adaptation. (iv) The counterfactual loss operates on clinically implausible perturbations but does not formally verify robustness against adversarial attackers; an ongoing work stream pairs HEAL-GTN with certified randomized smoothing. (v) Cox-based mortality heads assume proportional hazards; time-varying extensions such as DeepHit [65] could be integrated with modest refactoring.

VIII. CONCLUSION

We have presented HEAL-GTN, a hierarchical graph-temporal network that mines longitudinal multi-modal electronic health records to predict sepsis six hours ahead, acute kidney injury within 24 hours, and in-hospital mortality. The method couples modality-specific encoders with a heterogeneous patient-concept graph, propagates signal through three graph-attention layers and a bidirectional LSTM with temporal self-attention, and closes with three calibrated task heads trained under a differentially private federated protocol. On MIMIC-IV, HEAL-GTN attains AUROCs of 0.912, 0.889 and 0.905 for sepsis, AKI and mortality, respectively, improving on the strongest prior baseline by 2.9 to 4.0 percentage points while also producing better calibration and the highest decision-curve net benefit. A critical comparative study against 16 baselines confirms that graph reasoning, cross-modal fusion, temporal self-attention and calibrated uncertainty each contribute to performance. Future work will extend HEAL-GTN with certified randomized smoothing, continuous-time dynamic graphs, and culturally diverse external cohorts in partnership with the Malaysian Ministry of Health.

REPRODUCIBILITY

The source code, training scripts, federated-learning simulator and a synthetic-cohort generator are released at <https://github.com/sathishv/heal-gtn> under the MIT license. The repository contains dockerized dependencies, MIMIC-IV and eICU-CRD preprocessing pipelines compatible with the PhysioNet credentialed-access workflow, configuration files for all reported experiments, pretrained model checkpoints trained on the synthetic cohort, and evaluation notebooks.

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We thank the editor and both reviewers for their careful reading. Each comment is answered below; section, table, figure and equation numbers refer to the revised manuscript. In addition, every reference was checked against its published record. Author lists, titles, venues and page numbers were corrected where necessary, and entries that did not match the work being described were replaced by the correct publications.

Reviewer 1

- 1) *The first sentence of the abstract should not exceed 25 words.* The first sentence now has 17 words: “Electronic health records combine clinical notes, laboratory results, physiological signals, medication orders and diagnosis codes for each hospital admission.”
- 2) *Quantities above 10 should be written in Arabic numerals.* All quantities above 10 are now written as numerals throughout, for example 16 baselines, 24 hours, 10 to 15 per cent, more than 200 hospitals, 11 points and 15 bins.
- 3) *Order the abstract as background, research question, significance, methods and results.* The abstract now follows this order: background (sentences 1 and 2), research question (3), significance (4), methods (5 and 6) and results (7 and 8), followed by the availability statement.
- 4) *State the shortcomings of previous research in one sentence and remove subjective assertions.* Sentence 2 of the abstract states the shortcomings of previous research in one sentence. The phrase “among the richest and most difficult streaming data-mining substrates” and similar evaluative wording were removed.
- 5) *Delete the paragraph describing the structure of the paper.* Deleted.
- 6) *Add literature from 2023 onwards.* We added 14 works published from 2023 onwards [7], [19]–[21], [24], [26], [32], [33], [37], [43], [53], [55], [62], [64], in addition to the MIMIC-IV reference [27]. They are discussed in Sections I, II, IV and V.
- 7) *The problem definition in Section III is insufficient.* Section III was extended. It now defines the time-stamped information set available at prediction time (1) and the resulting admissibility condition, the node-type and edge-type maps of the heterogeneous graph, the three prediction targets with their horizons and the censoring of the mortality outcome, the link to the training objective (5), the federated constraint with the formal definition of differential privacy (2), and the evaluation measures.
- 8) *Merge Section VIII into Section VII or delete it.* The former Sections VII and VIII are merged into Section VII, “Discussion, Deployment Considerations and Broader Impact,” which now also contains the limitations.

Reviewer 2

- 1) *Risk of temporal leakage from diagnosis codes; unclear availability of notes, medications and laboratory features.* We added Section V-B, which states for each input whether it is time-stamped and available during the stay. Vital signs, laboratory values, medication administrations and notes are time-stamped; ICD-10 diagnosis codes in MIMIC-IV and eICU-CRD are assigned at or after discharge. The section states that, in the MIMIC-IV and eICU-CRD experiments, these codes enter the graph and the sepsis label is derived from them, so those results describe admission-level risk stratification, and that PhysioNet 2019, which contains no diagnosis codes, provides a check of the sepsis head without this source of leakage. The prediction horizons are now defined formally through (1) in Section III, and the issue is listed as limitation (ii) in Section VII.
- 2) *Inconsistent cohort statistics; incomplete outcome and exclusion definitions.* All counts in Table II are now ICU stays, consistent with the text, and the table caption says so; for PhysioNet 2019 each patient contributes one stay. Section V-A now gives the inclusion and exclusion criteria for each dataset, the ICD-10 codes defining the sepsis label, the PhysioNet 2019 label definition, the KDIGO creatinine criterion with its baseline for AKI, and the definition of in-hospital mortality with censoring at discharge.
- 3) *Whether test patients or test-set statistics influence the graph or training.* Sections IV-A and V-C now state that each patient’s graph contains only that patient’s own concepts, so no test stay appears as a node in any training graph; that the concept embeddings, relation weights and edge priors are fitted on the training split only; and that the validation split alone is used for early stopping, hyperparameter search and the isotonic calibrator. We also state that the split is made by stay.
- 4) *Insufficient information to reproduce the privacy budget; overstated privacy claims.* Section IV-G now reports every setting the accountant uses: the mechanism and library, clipping norm, noise multiplier, batch size and sampling rate, local epochs, number of rounds, participation rate, the composition over all local steps and rounds, the value of δ relative to the client dataset sizes, and the unit of privacy. We reworded two statements that overstated the guarantee:

the acceptability of the budget under the Malaysian Personal Data Protection Act is now presented as a matter for the data controller to assess, and the model-inversion result is now described as the outcome of the attack evaluated rather than as a formal bound.

- 5) *AUPRC, F1 and decision-curve metrics are claimed but the main table reports mostly AUROC and ECE.* Section V-E now states where each metric is reported: Table III gives AUROC for all three tasks and the ECE for AKI, Fig. 5(b) gives the sepsis precision-recall curves, Fig. 6 gives the reliability diagrams and decision curves, and Section VI-G reports counterfactual robustness.